



Lecture Notes

Institute Name:	Silver Oak College of Computer Application		
Programme Name:	Integrated Master Of Science Information Technology	Semester:	7th
Course Code:	MDC5011C	Course Name:	Ethical and Legal Frameworks for Digital Systems
Unit No.:	2	Unit Name:	Fundamentals of Cyber Laws and Privacy

1. Introduction to Cyber Law

Cyber law refers to the legal rules and regulations that govern activities involving computers, digital systems, networks, the internet, electronic communications, and digital information.

Traditional laws were primarily designed for physical activities. The growth of the internet introduced new situations such as:

- Online fraud
- Identity theft
- Cyberstalking
- Unauthorized access
- Data breaches
- Online harassment
- Digital piracy
- Electronic contracts
- Electronic payments
- Misuse of personal information

Cyber law provides a legal framework for addressing these activities and establishing **rights, responsibilities, offences, penalties, and legal procedures.**

Objectives of Cyber Law

Cyber laws generally aim to:

1. Recognize electronic transactions and records.
2. Prevent and address cybercrime.
3. Protect digital information.
4. Establish legal validity for electronic authentication.
5. Protect users and organizations from digital misuse.
6. Define responsibilities of digital service providers and users.



7. Support secure electronic commerce and communication.
8. Provide mechanisms for investigation and enforcement.

2. Need for Cyber Laws in Digital Society

Modern society increasingly depends on digital technologies for:

- Banking
- Education
- Healthcare
- Government services
- E-commerce
- Social networking
- Digital payments
- Business communication
- Cloud computing
- Artificial intelligence

This dependence creates legal and ethical challenges.

Example

A person creates a fake social-media account using another person's photograph and uses it to deceive others.

Several issues may arise:

Identity misuse → Privacy violation → Online fraud → Evidence collection → Legal investigation

Cyber laws provide mechanisms for dealing with such digital activities.

Importance of Cyber Laws

1. Protection from Cybercrime

Cyber laws help define unlawful digital activities and their legal consequences.

2. Legal Recognition of Electronic Transactions

Electronic records and electronic authentication are essential for digital commerce and government services.

3. Protection of Digital Information



Organizations process large quantities of personal and confidential information. Legal frameworks establish obligations relating to its handling.

4. Support for Digital Commerce

E-commerce depends on legally recognized electronic contracts, records, authentication, and transactions.

5. Privacy Protection

Digital technologies enable large-scale collection and processing of personal information. Privacy laws help establish boundaries around such processing.

6. Accountability

Cyber laws help determine who may be legally responsible when digital systems or information are misused.

3. Information Technology Act, 2000

In India, the **Information Technology Act, 2000 (IT Act 2000)** is a major legislation governing electronic records, electronic communication, electronic commerce, and several cyber-related offences.

The Act was enacted to provide legal recognition to electronic transactions and facilitate electronic filing and related digital activities.

It also contains provisions dealing with various computer-related offences and contraventions.

Major Areas Covered

The IT Act addresses areas such as:

- Electronic records.
- Electronic authentication.
- Digital signatures.
- Electronic governance.
- Cyber offences.
- Unauthorized access and damage to computer systems.
- Intermediary-related provisions.
- Data and information protection obligations.
- Investigation and adjudication of certain digital offences.



4. Important Concepts under the IT Act

4.1 Electronic Record

An **electronic record** refers broadly to information created, stored, received, or transmitted in electronic form.

Examples:

- PDF documents
- Emails
- Electronic invoices
- Digital applications
- Electronic agreements
- Online transaction records

Electronic records are fundamental to digital governance and electronic commerce.

4.2 Electronic Signature

An electronic signature is an electronic method used to authenticate or sign an electronic record.

It establishes a relationship between:

Signer → Electronic authentication → Electronic record

4.3 Digital Signature

A **digital signature** is a cryptographic mechanism used to authenticate an electronic document and help ensure its integrity.

It generally uses:

- Private key
- Public key
- Digital certificate
- Cryptographic algorithms

Basic Process

Document → Hash → Encryption using private key → Digital Signature

The recipient can use the corresponding public key to verify the signature.

Benefits



- Authentication
- Integrity
- Non-repudiation support
- Secure electronic transactions

5. Common Cyber Crimes and Legal Implications

Cybercrime refers to unlawful activity involving computers, networks, digital devices, or digital information.

5.1 Unauthorized Access

Accessing a computer system, account, or network without authorization.

Example:

An employee uses another employee's credentials to access restricted files.

Possible consequences

- Legal action
- Financial penalties
- Organizational disciplinary action
- Civil or criminal liability depending on the circumstances and applicable law

5.2 Data Theft

Unauthorized copying, extraction, or misuse of digital information.

Examples:

- Customer database theft
- Password theft
- Intellectual property theft
- Financial information theft

5.3 Identity Theft

Using another person's identifying information without authorization.

Examples:

- Misusing another person's credentials.
- Creating fraudulent accounts.
- Using stolen identity information for financial transactions.



5.4 Phishing

Phishing uses deceptive messages or websites to trick users into revealing credentials, financial information, or other sensitive data.

Example

An attacker sends:

"Your bank account will be blocked. Click this link to verify your account."

The link leads to a fraudulent website designed to capture login information.

5.5 Cyberstalking

Repeatedly using digital communication or online platforms to harass, threaten, monitor, or intimidate another person.

5.6 Online Harassment

Unwanted, abusive, threatening, or harmful behavior conducted through digital platforms.

Examples:

- Threatening messages
- Repeated abusive communication
- Online intimidation
- Coordinated harassment

5.7 Cyber Defamation

Publishing false or damaging statements about an individual or organization through digital platforms may create legal consequences depending on the circumstances and applicable laws.

5.8 Ransomware

Malicious software that prevents access to files or systems, commonly by encrypting data, and demands payment.

Possible consequences include:

- Financial loss
- Operational disruption
- Data loss
- Privacy violations



- Recovery expenses
- Regulatory consequences where applicable

6. Cyber Crime — General Legal Process

A cyber incident may involve several stages:

Incident → Evidence preservation → Investigation → Identification of accused → Legal proceedings → Penalty/remedy

Digital evidence may include:

- Emails
- Log files
- IP-related records
- System files
- Transaction records
- Chat messages
- Digital photographs
- Device information

Importance of Digital Evidence

Digital evidence must be collected and handled carefully because improper handling can affect its reliability and legal admissibility.

7. Digital Signatures and Electronic Records

7.1 Traditional Signature vs Digital Signature

Traditional Signature	Digital Signature
Usually handwritten	Cryptographically generated
Physically attached to document	Associated electronically with a record
Verification may depend on handwriting/signature examination	Verification uses cryptographic mechanisms
Physical document commonly involved	Electronic record involved

7.2 Properties of a Digital Signature



Authentication

Helps verify the identity associated with the signature.

Integrity

Helps detect whether the signed data has been modified.

Non-repudiation

Provides evidence supporting the association of the signer with the signed transaction, subject to the applicable legal and technical framework.

7.3 Electronic Records

Electronic records can include:

- Email
- Digital invoices
- Online applications
- Electronic contracts
- E-tickets
- Digital certificates
- Online transaction records

The legal recognition of electronic records is essential for a digital economy.

8. Introduction to Privacy

Privacy is the ability of individuals to have appropriate control over information relating to themselves and to expect that personal information will be handled appropriately.

Privacy becomes particularly important in digital environments because organizations can collect, store, analyze, share, and retain enormous quantities of personal data.

Examples of Personal Information

- Name
- Address
- Telephone number
- Email address
- Identification information
- Online identifiers
- Financial information

- Location information
- Digital account information

9. Privacy vs Data Security

Privacy and security are related but different concepts.

Privacy	Data Security
Concerned with appropriate collection and use of personal data	Concerned with protecting data and systems
Focuses strongly on individual rights and appropriate processing	Focuses strongly on confidentiality, integrity and availability
Addresses purpose, transparency and user rights	Addresses technical and organizational safeguards
Includes responsible data practices	Includes encryption, access control, authentication, etc.

Example

An organization may securely encrypt a customer's information.

This addresses **data security**.

But if the organization collected that information for one purpose and later uses it for an unrelated purpose without an appropriate legal basis or authorization, a **privacy issue** may still exist.

10. Personal Data and Sensitive Information

10.1 Personal Data

Personal data generally means information relating to an identifiable individual.

Examples:

- Name
- Mobile number
- Email address
- Identification information
- Online identifier
- Location information



The exact legal definition depends on the applicable law.

10.2 Sensitive Information

Some categories of information can create greater risks if exposed or misused.

Examples may include:

- Financial information
- Health-related information
- Authentication credentials
- Biometric information
- Precise location information
- Other information that can create significant privacy or security risks

Important: Legal definitions of "sensitive personal data" differ between laws and jurisdictions. Therefore, students should distinguish between the **general concept of sensitive information** and a specific statutory definition.

11. Digital Personal Data Protection Concepts

India's **Digital Personal Data Protection Act, 2023 (DPDP Act)** establishes a framework for processing digital personal data and protecting individuals' privacy.

The Act uses important concepts that students should understand.

11.1 Data Principal

The **Data Principal** is the individual to whom the personal data relates.

Example

If a university stores information about a student:

Student = Data Principal

11.2 Data Fiduciary

A **Data Fiduciary** is an entity that determines the purpose and means of processing personal data.

Example

If an organization decides:



- Why student data is collected.
- What data is collected.
- How the data will be used.

The organization may act as a **Data Fiduciary**.

11.3 Data Processor

A **Data Processor** processes personal data on behalf of a Data Fiduciary.

Example

A university may use an external cloud service to process or store student information under contractual arrangements.

The university may be the **Data Fiduciary**, while the service provider may act as a **Data Processor**, depending on the specific arrangement.

12. Key Concepts of the DPDP Framework

Consent

Consent is an important mechanism through which individuals authorize processing of their personal data where consent is the applicable basis.

Consent should be:

- Free
- Specific
- Informed
- Unambiguous
- Capable of being withdrawn

Notice

Individuals should receive appropriate information about how their personal data is processed.

A privacy notice can communicate:

- What information is collected.
- Purpose of processing.
- How the information is used.
- Relevant rights.
- Contact or grievance mechanisms where applicable.



Purpose Limitation

Personal data should be processed for legitimate purposes associated with the applicable legal framework.

Data Minimization

Organizations should avoid unnecessary collection of personal information.

Example

A library system may need:

Student ID + Name + Book information

It generally does not need unrelated personal information simply because it can technically collect it.

Security Safeguards

Organizations should implement reasonable safeguards to protect personal data from:

- Unauthorized access
- Data breaches
- Accidental loss
- Unauthorized disclosure
- Improper modification

13. Rights of Digital Users

Digital users have important rights and responsibilities in the digital environment.

Under applicable privacy and digital laws, individuals may have rights relating to their personal information.

Major Rights and Concepts

1. Right to Information

Users should be provided appropriate information about the processing of their personal data.

2. Right to Access



Depending on the applicable legal framework, individuals may have rights to obtain information about their personal data and its processing.

3. Right to Correction

Individuals may have mechanisms to request correction of inaccurate personal information where provided by law.

4. Right to Erasure

Certain privacy frameworks provide rights to request deletion or erasure of personal data under specified conditions.

5. Right to Withdraw Consent

Where processing is based on consent, individuals may have the ability to withdraw consent, subject to legal requirements and consequences.

6. Right to Grievance Redressal

Users should have mechanisms for raising concerns regarding improper handling of personal information.

14. Responsibilities of Digital Users

Digital rights come with corresponding responsibilities.

Users should:

- Protect their login credentials.
- Use strong passwords.
- Enable MFA.
- Avoid sharing sensitive information unnecessarily.
- Read important privacy notices.
- Verify websites before entering personal information.
- Report suspicious activities.
- Respect other people's privacy.
- Avoid unauthorized access.
- Avoid spreading private information.
- Respect intellectual property.
- Use digital platforms responsibly.
- Follow applicable organizational policies.

15. Privacy by Design

Privacy by Design means considering privacy requirements during the design and development of a technology system rather than attempting to add privacy protections after the system is completed.

Traditional Approach

Develop System → Collect Data → Identify Privacy Problems → Fix Problems

Privacy-by-Design Approach

Identify Privacy Requirements → Design Privacy Controls → Develop System → Test → Monitor

Examples

- Collect minimum necessary information.
- Use appropriate access controls.
- Provide clear privacy notices.
- Encrypt sensitive information.
- Establish retention and deletion mechanisms.
- Provide user controls over personal information.

16. Cyber Law, Privacy and Cybersecurity Relationship

These three areas are closely connected.

Cyber Law

Defines **legal requirements and prohibited activities**.

Privacy

Focuses on **appropriate handling and protection of personal information**.

Cybersecurity

Provides **technical and organizational mechanisms for protecting systems and information**.

A simple relationship is:

Cyber Law → Legal Framework

Privacy → Individual/Data Protection



Cybersecurity → Technical & Organizational Protection

Together:

Law + Privacy + Security = Responsible Digital Environment

17. Case Study: Phishing and Personal Data Theft

Situation

A student receives an email claiming to be from a university. The email requests the student's username, password, Aadhaar-related information, and bank details.

The student enters the information into a fake website.

Issues

Cybersecurity issue:

Credentials were stolen.

Privacy issue:

Personal information was disclosed to an unauthorized party.

Cybercrime issue:

The attacker may have committed offences involving unauthorized access, cheating, identity-related misuse, or other offences depending on the facts and applicable law.

Responsible Response

1. Stop interacting with the fraudulent website.
2. Change compromised passwords.
3. Enable MFA.
4. Inform the university/organization.
5. Contact relevant financial institutions if financial information was exposed.
6. Preserve relevant evidence.
7. Report the incident through appropriate channels.

18. Case Study: Unauthorized Use of Student Data

Suppose a university collects student information for academic administration. An employee downloads the database and shares it with an unauthorized third party.

Potential concerns



- Unauthorized access.
- Unauthorized disclosure.
- Privacy violation.
- Breach of organizational policy.
- Potential legal and regulatory consequences.

Preventive Measures

Access control + Authentication + Data minimization + Monitoring + Employee awareness + Incident response

19. Cyber Ethics and Legal Compliance

A behavior can be technically possible but still be **unethical or illegal**.

Example

A person discovers a vulnerability in a university website.

Three possible actions:

A. Exploit it for personal benefit

→ Potentially illegal and unethical.

B. Ignore it

→ Does not help the organization address the vulnerability.

C. Responsibly report it to the authorized security team

→ Responsible cybersecurity behavior.

This demonstrates the importance of **responsible disclosure and ethical technology use**.